DONOR DONATION ANALYSIS USING SQL

SQL capstone portfolio

BY

UJAM, PRECIOUS MMESOMA

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Database: PostgreSQL 18

Contents

EXECUTIVE SUMMARY

This project analyzes a donor donation dataset using PostgreSQL to identify donation patterns and generate actionable insights that can support fundraising strategies. The analysis focused on understanding donor behaviour by gender, donation frequency, job field, state, and donation value. Data quality checks were performed to identify missing values, duplicates, and inconsistencies before conducting the analysis.

Key findings indicate that donation amounts vary across states, occupations, and donation frequencies. Certain states contribute significantly more donations than others, while donors with specific giving frequencies exhibit higher average donation values. The analysis also identifies the highest donating individuals within each state and explores relationships between donor characteristics and donation behaviour.

Based on these findings, recommendations are provided to help the organization improve donor retention, increase donation frequency, attract new donors, and maximize fundraising revenue. The project demonstrates the use of SQL for data exploration, data cleaning, aggregation, joins, Common Table Expressions (CTEs), window functions, and view creation.

BUSINESS CONTEXT AND PROBLEM STATEMENT

Business Context:

The organization maintains information about donors and their donations. Understanding donor behaviour is essential for improving fundraising campaigns, identifying valuable donor segments, and increasing total donations.

Business Problem:

The organization seeks to find clear insights into:

- Overall donation performance.
- Donation behaviour by gender.
- Donation behaviour by state.
- Donation frequency.
- Occupation of donors.
- Characteristics of high-value donors

Objectives:

- Increase the number of donors in the database.
- Increase the donation frequency of the donors.
- Increase the value of donations in the database.

This analysis aims to answer these questions using SQL, and inform business decisions.

DATA OVERVIEW & SCHEMA

The dataset consists of two related tables: Donation Data and Donor Data. Together, they contain demographic, behavioural, and donation information for 1,000 donors. The tables are linked using the id field, which uniquely identifies each donor. The data was used to analyze donation patterns across different demographic and behavioural characteristics.

Table Name	Description	Number of Records
Donation Data	Contains donor demographic details and donation amounts	1000
Donor Data	Contains donor lifestyle and preference information	1000

Donation Data Table:

	id integer	first_name character varying (50)	last_name character varying (50)	email character varying (50)	gender character varying (50)	job_field character varying (50)	donation integer	state character varying (50)	shirt_size character varying (50)
1	1	Nefen	Rogerson	nrogerson0@tumblr.com	Male	Human Resources	28	Colorado	3XL
2	2	Kippar	Saffin	ksaffin1@si.edu	Male	Human Resources	292	California	2XL
3	3	Panchito	Crichley	pcrichley2@wired.com	Female	Engineering	178	California	XL
4	4	Skippy	McTavy	smctavy3@friendfeed.com	Male	Sales	304	Illinois	S
5	5	Trudie	Codner	tcodner4@newyorker.com	Male	Business Development	219	Florida	M

Donor Data Table:

	id integer	donation_frequency character varying (100)	university character varying (100)	car character varying (100)	second_language character varying (100)	favourite_colour character varying (100)	movie_genre character varying (100)
1	1	Daily	Ecole Européen des Affaires	Toyota	Kyrgyz	Green	Comedy|Documentary
2	2	Yearly	University of Calicut	Chevrolet	[null]	Teal	Drama
3	3	Weekly	Central Ostrobothnia University of Applied Sciences	Suzuki	[null]	Purple	Action|Adventure|Thriller
4	4	Weekly	Universidad Autónoma del Estado de Hidalgo	Ford	Georgian	Maroon	Action|Comedy|Crime|Dr
5	5	Yearly	[null]	Honda	[null]	Yellow	Romance

DATA DICTIONARY

Donation Data:

Id - Unique donor identifier.

First_name – Donor's first name

Last_name – Donor's last name

Email – Donor's email address

Gender - Donor gender

Job_field – Donor occupation

Donation - Amount donated.

State - State of residence

Shirt_size – Donor's shirt size (XS, S, M, L, XL, etc.)

Donor Data:

Id – Unique donor identifier

Donation_frequency - Frequency of donations

University – University attended

Car - Vehicle owned by donor

Second_language – Second language of donor

Favourite_colour – Favourite colour of donor

Movie_genre – Favourite movie genre of donor

Assumptions

- Each ID represents one donor.
- Donation values are recorded in US Dollars.
- Missing university information does not affect donation analysis.
- Missing second_language values do not affect business conclusions.

DATA QUALITY ASSESSMENT

Missing Values:

Missing values were identified in:

- University
- Second_language

These variables were not required for the business questions and were therefore retained without imputation.

Duplicate Records:

No duplicate donor IDs were identified.

Referential Integrity:

All donation records matched corresponding donor records using the id field.

METHODOLOGY

The analysis followed these steps:

1. Dataset queries were loaded into PostgreSQL to create the tables.
2. Exploratory data analysis was performed to gain insight on total number of donors, total number of donations made, distinct genders, distinct frequencies distinct states, donation amount range.
3. Data quality was assessed to check for missing values, duplicated records, orphan donations.
4. Datasets were joined where necessary using the unique donor ID.
5. SQL aggregation functions (SUM, COUNT, AVG) were used.
6. GROUP BY for segmentation was used.
7. Common Table Expressions (CTEs) were used for readability.
8. Window functions (RANK()) were used to rank donors, and identify the highest donor in each state.
9. Summary views were created for reporting.

ANALYSIS & FINDINGS

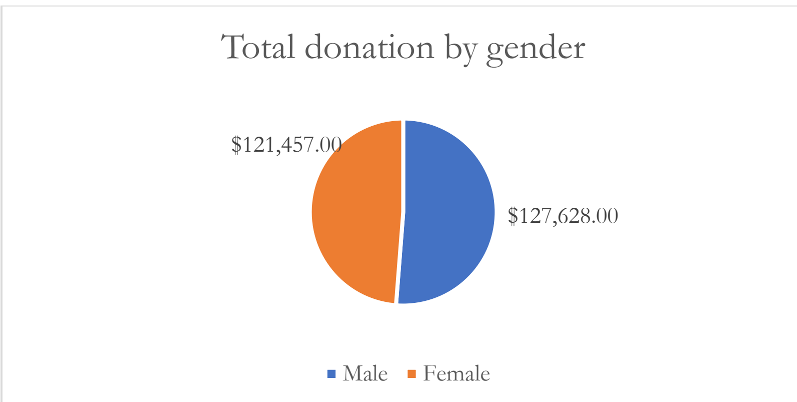
Business Question 1: How Much Is the Total Donation?

Total donation
$ 249,085.00

Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

Total donation received from all donors was $ 249,085.00. This figure represents the overall funds generated from the 1,000 donors in the dataset and serves as the baseline for evaluating donation performance across different donor groups.

Business Question 2: What Is the Total Donation by Gender?



Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

The results show the total amount donated by each gender. Comparing these totals helps identify which gender contributed more overall and can guide targeted fundraising campaigns and donor engagement strategies.

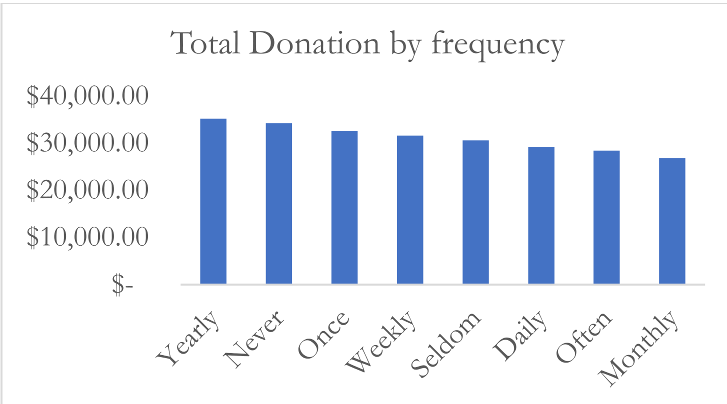
Business Question 3: Show the Total Donations and Number of Donations by Gender

Gender	Total donation	No. of donations
Male	$ 127,628.00	492
Female	$ 121,457.00	508

Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

This analysis compares both the number of donations and the total amount donated by each gender. It helps determine whether higher total donations are driven by a larger number of donors or by higher average donation amounts.

Business Question 4: What Is the Total Donation made by frequency of donation?



Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

This analysis shows how much each donation frequency category contributed to the total donations. It provides insight into whether frequent donors or occasional donors generate more revenue for the organization.

Business Question 5: What Is the Total Donation and Number of Donations by Job Field?

Human Resources	$	23,060.00	93
Research and Development	$	22,862.00	84
Product Management	$	22,798.00	90
Business Development	$	22,266.00	94
Engineering	$	21,968.00	93
Training	$	21,721.00	84
Accounting	$	20,504.00	80
Services	$	19,858.00	80
Support	$	19,475.00	79
Sales	$	19,009.00	83
Marketing	$	18,255.00	74
Legal	$	17,309.00	66

Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

This analysis identifies which occupations contribute the highest donation amounts and have the largest donor counts. The findings can help the organization target professions with stronger giving potential.

Business Question 6: What Is the Total Donation and Number of Donations Above $200?

Total donation	No. of donations
$ 205,892.00	586

Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

This analysis focuses on higher-value donations exceeding $200. It measures the contribution of larger donations to total fundraising income and helps identify the importance of high-value donors.

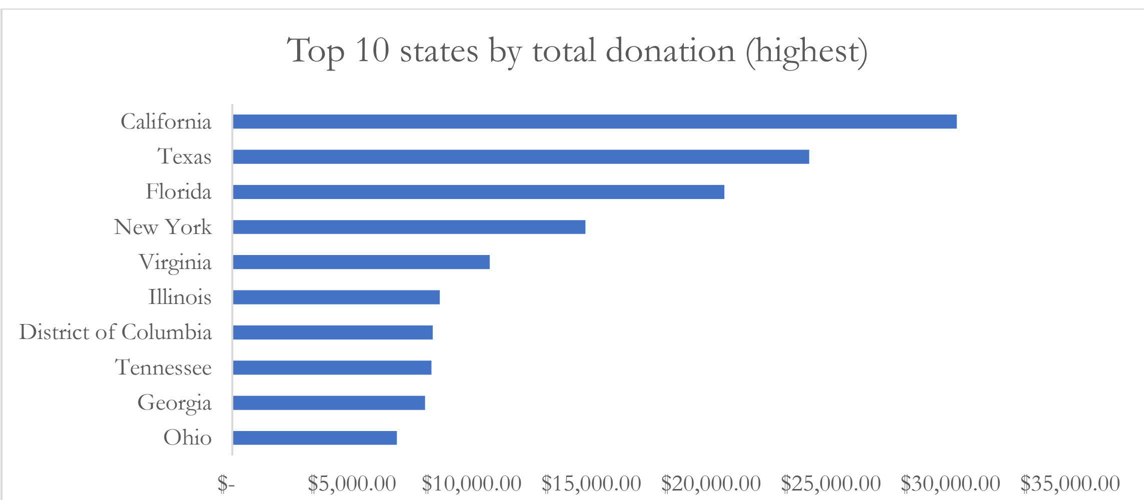
Business Question 7: What Is the Total Donation and Number of Donations Below $200?

Total donation	No. of donations
$ 42,593.00	411

Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

This analysis summarizes donations below $200. Comparing these results with donations above $200 helps assess whether fundraising relies more on many smaller donations or fewer larger contributions.

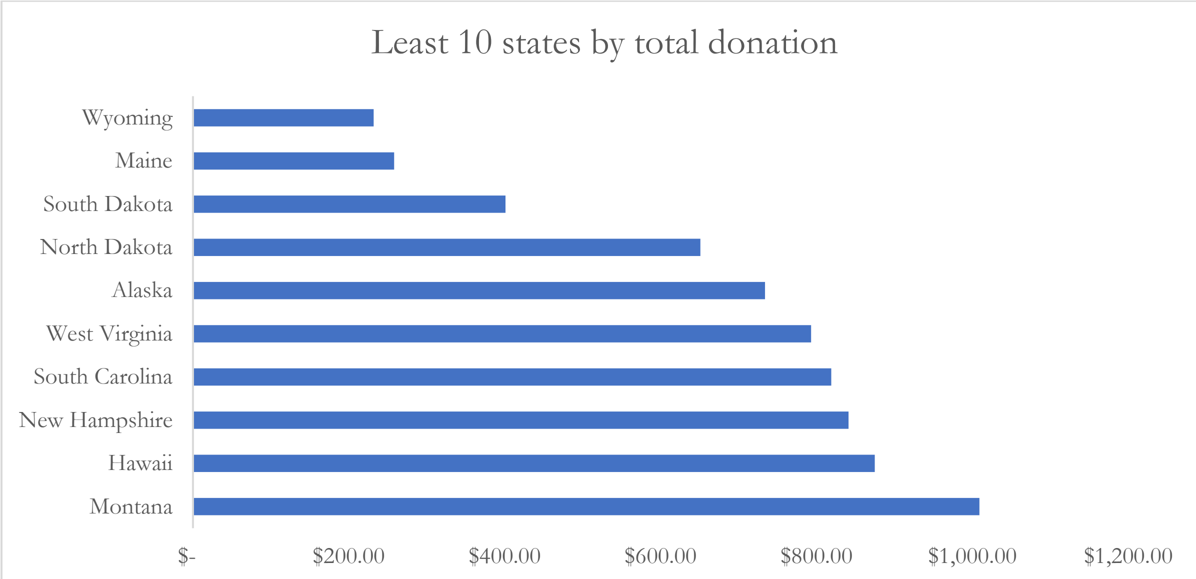
Business Question 8: Which Top 10 States Contribute the Highest Donations?



Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

The top ten states generated the highest total donation amounts. These states represent key fundraising regions where future campaigns and donor engagement efforts may produce the greatest return.

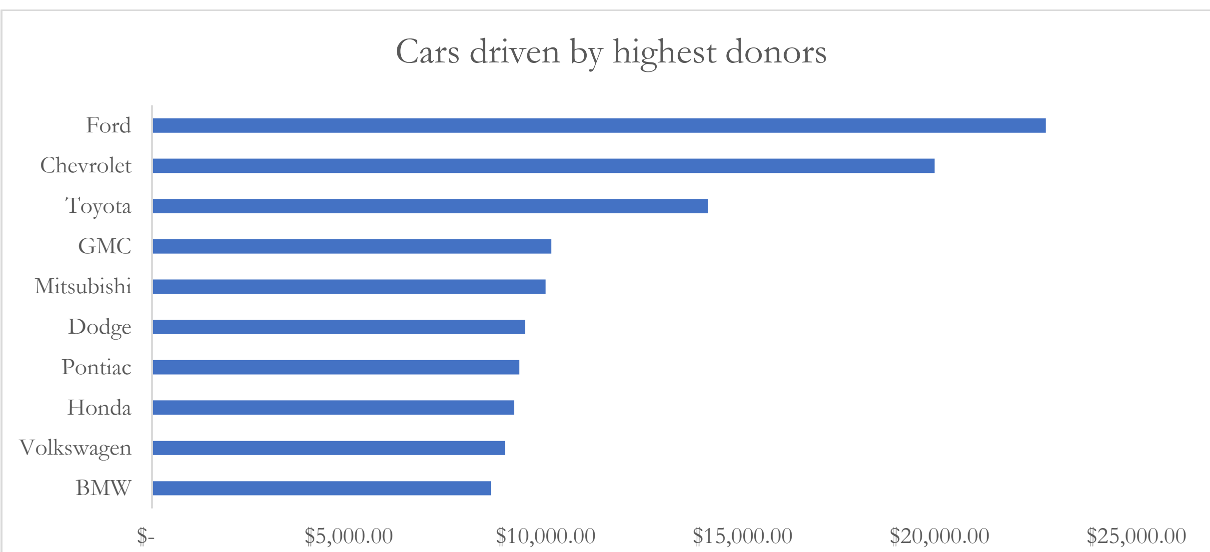
Business Question 9: Which Top 10 States Contribute the Least Donations?



Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

The states with the lowest total donations may represent underserved regions or areas with lower donor participation. These findings can help the organization identify locations where awareness campaigns or fundraising initiatives could be strengthened.

Business Question 10: What Are the Top 10 Cars Driven by the Highest Donors?



Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

This analysis explores the relationship between donors' vehicle preferences and donation behaviour. Although car ownership does not directly influence giving, it provides an additional donor characteristic that may be useful for segmentation and marketing purposes.

Additional Analyses:

Business Question 11: What Is the Average Donation by Donation Frequency?

Donation frequency	No. of donations	Total donation	Average donation amount
Yearly	135	$ 35,266.00	261.23
Seldom	118	$ 30,650.00	259.75
Never	133	$ 34,263.00	257.62
Often	111	$ 28,476.00	256.54
Once	128	$ 32,666.00	255.2
Weekly	129	$ 31,645.00	245.31
Daily	128	$ 29,249.00	228.51
Monthly	118	$ 26,870.00	227.71

Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

The average donation amount varies across donation frequency groups. This analysis helps determine whether regular donors also contribute larger amounts on average or whether less frequent donors tend to make higher-value donations.

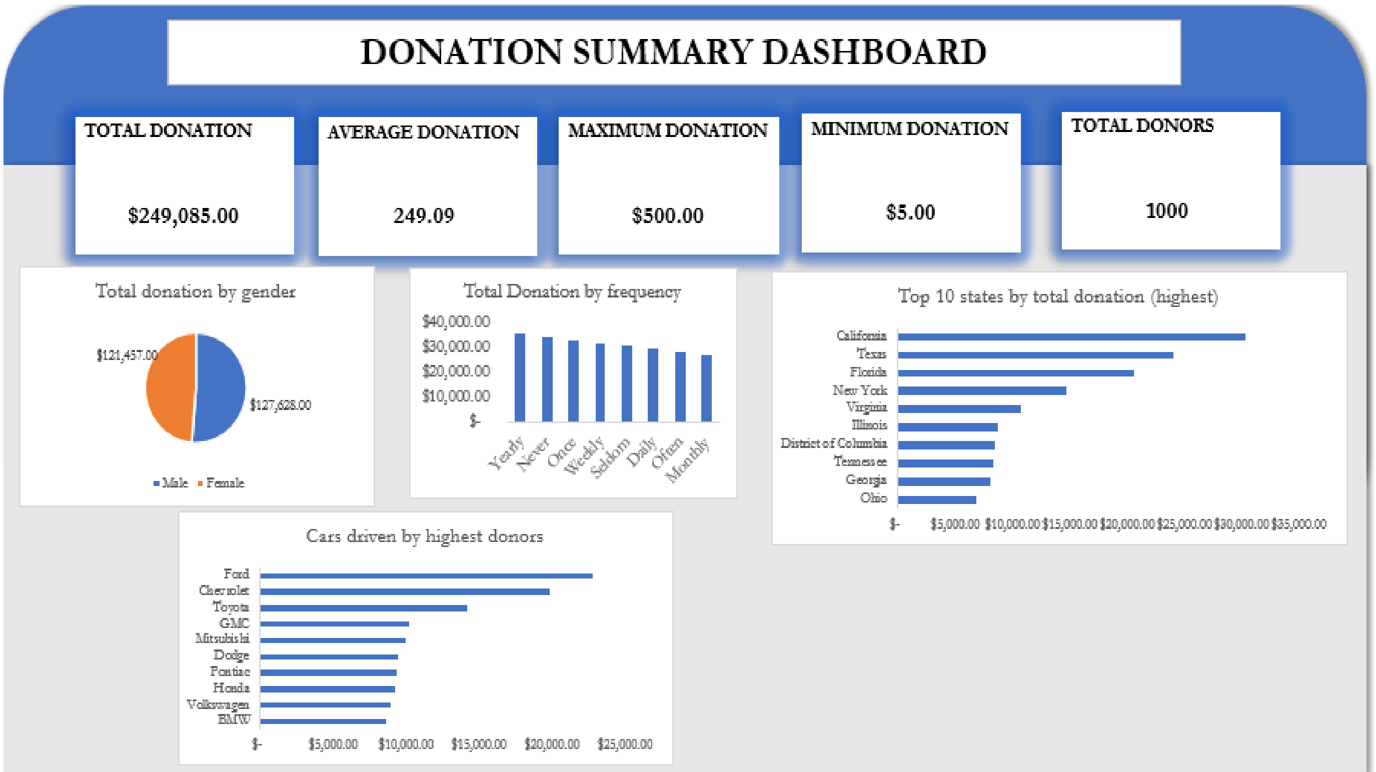
Business Question 12: Top 20 donors?

ID	First name	Last name	Donation amount
264	Wallie	Leather	$ 500.00
139	Beverlie	Andriesse	$ 500.00
769	Peder	Rilton	$ 499.00
35	Clevie	Camilletti	$ 499.00
480	Worthy	Le feaver	$ 498.00
965	Amalea	Knill	$ 497.00
969	Nathaniel	McGenn	$ 494.00
76	Tonnie	Stockney	$ 494.00
500	Corbett	Lansdale	$ 494.00
941	Corbin	Rawne	$ 493.00
565	Beverlee	Camacke	$ 493.00
290	Hurley	Bogey	$ 492.00
729	Eddi	Atcherley	$ 492.00
153	Emmit	McKenzie	$ 491.00
296	Babbette	Fyers	$ 491.00
765	Karilynn	Ivan	$ 490.00
203	Ludvig	Alexsandrowicz	$ 489.00
90	Charlotta	Bellison	$ 489.00
561	Broderick	Dimitrijevic	$ 489.00
931	Padraig	Trittam	$ 488.00

Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

Using a window function (RANK()), the highest donor in each state was identified. This analysis highlights major contributors across different regions and may support personalized donor recognition and retention strategies.

Business Question 13: Overall Donation Summary Dashboard



Source: Author's analysis using the supplied donation and donor datasets in PostgreSQL.

KEY INSIGHTS

- Most donations originated from a small number of high-performing states.
- Donation frequency influences average donation value.
- The total donation made by donors who donated yearly, was the highest.
- Monthly donors generated the least total donation.
- There were more female donors than male donors.
- Total amount donated by the male donors was higher than the female total donation.
- Certain job fields such as human resources, research & development, product management, business development contributed substantially more donations than others.
- High-value donors were concentrated in specific geographic regions such as California, Texas, Florida, New York, Virginia.
- Donor characteristics can be used to support targeted fundraising campaigns.

RECOMMENDATIONS

Recommendation 1:

We recommend focusing fundraising campaigns on the highest-contributing states because they generated the greatest total donations. This is expected to increase fundraising efficiency.

Recommendation 2:

We recommend encouraging occasional donors to become monthly donors through personalized communication because more frequent donors generally contribute greater overall value.

Recommendation 3:

We recommend implementing targeted engagement strategies for high-value occupations identified in the analysis because these donor groups contributed substantial donation totals.

Recommendation 4:

We recommend recognizing and retaining top donors through loyalty programmes because retaining existing high-value donors is often more cost-effective than acquiring new donors.

Recommendation 5:

We recommend improving the completeness of donor profile information by reducing missing values in non-financial attributes, enabling richer future analyses and more effective donor segmentation.

LIMITATIONS & ASSUMPTIONS

- The dataset represents a single snapshot in time, so trends over time cannot be assessed.
- Donation amounts are assumed to be accurate and measured consistently.
- Missing values in some donor profile fields may limit certain analyses.
- The dataset does not include campaign information or donation dates, preventing time-series analysis.

APPENDIX

SQL Queries:

```
/* Question 1: Total Donation? */
SELECT
   SUM(donation) AS total_donation
FROM donation_data;

/* Question 2: Total Donation by Gender? */
SELECT
   gender,
          SUM(donation) AS total_donation
FROM donation_data
GROUP BY gender
ORDER BY total_donation DESC;

/* Question 3: Total donations and number of donations by gender */
WITH gender_donation_summary AS (
   SELECT
             gender,
                    donation
          FROM donation_data
)
SELECT
   gender,
          SUM(donation) AS total_donation,
          COUNT(donation) AS number_of_donations
FROM gender_donation_summary
GROUP BY gender
ORDER BY total_donation DESC;
```

```
/* Question 4: Total donation made by frequency of donation */
WITH donation_frequency_summary AS (
   SELECT
      donor_data.donation_frequency,
             donation_data.donation
   FROM donor_data
   INNER JOIN donation_data
   ON donor_data.id = donation_data.id
)
SELECT
   donation_frequency,
          SUM(donation) AS total_donation
FROM donation_frequency_summary
GROUP BY donation_frequency
ORDER BY total_donation DESC;

/* Question 5: Total donations and number of donations by job field */
WITH job_field_donation_summary AS (
   SELECT
             job_field,
                    donation
          FROM donation_data
)
SELECT
   job_field,
          SUM(donation) AS total_donation,
          COUNT(donation) AS number_of_donations
FROM job_field_donation_summary
```

```
GROUP BY job_field
ORDER BY total_donation DESC;

/* Question 6: Total donation and number of donations above $200 */
WITH donations_above_200 AS (
   SELECT
             donation
         FROM donation_data
         WHERE donation > 200
)
SELECT
   SUM(donation) AS total_donation,
         COUNT(*) AS number_of_donations
FROM donations_above_200;

/* Question 7: Total donation and number of donations below $200 */
WITH donations_below_200 AS (
   SELECT
             donation
         FROM donation_data
         WHERE donation < 200
)
SELECT
   SUM(donation) AS total_donation,
         COUNT(*) AS number_of_donations
FROM donations_below_200;

/* Question 8: Top 10 states with the highest donations */
WITH state_donation_summary AS (
```

```
    SELECT
            state,
            donation
        FROM donation_data
)
SELECT
    state,
        SUM(donation) AS total_donation
FROM state_donation_summary
GROUP BY state
ORDER BY total_donation DESC
LIMIT 10;

/* Question 9: Top 10 states with the least donations */
WITH state_donation_summary AS (
    SELECT
            state,
                donation
        FROM donation_data
)
SELECT
    state,
        SUM(donation) AS total_donation
FROM state_donation_summary
GROUP BY state
ORDER BY total_donation ASC
LIMIT 10;

/*Question 10: Top 10 cars driven by the highest donors */
```

```
WITH donors_cars_table AS (
    SELECT
              donor_data.car,
              donation_data.donation
          FROM donor_data
          INNER JOIN donation_data
          ON donor_data.id = donation_data.id
)
SELECT
    car,
          SUM(donation) AS total_donation,
          COUNT(*) AS number_of_donors
FROM donors_cars_table
GROUP BY car
ORDER BY total_donation DESC
LIMIT 10;
```

SQL Query Screenshots:

Total Donation:

```
SELECT
    SUM(donation) AS total_donation
FROM donation_data;
```

Total Donation by Gender:

```
SELECT
    gender,
    SUM(donation) AS total_donation
FROM donation_data
GROUP BY gender
ORDER BY total_donation DESC;
```

Total donations and number of donations by gender:

```
WITH gender_donation_summary AS (
    SELECT
        gender,
        donation
    FROM donation_data
)
SELECT
    gender,
    SUM(donation) AS total_donation,
    COUNT(donation) AS number_of_donations
FROM gender_donation_summary
GROUP BY gender
ORDER BY total_donation DESC;
```

Total donation made by frequency of donation:

```
WITH donation_frequency_summary AS (
    SELECT
        donor_data.donation_frequency,
        donation_data.donation
    FROM donor_data
    INNER JOIN donation_data
    ON donor_data.id = donation_data.id
)
SELECT
    donation_frequency,
    SUM(donation) AS total_donation
FROM donation_frequency_summary
GROUP BY donation_frequency
ORDER BY total_donation DESC;
```

Total donations and number of donations by job field:

```
WITH job_field_donation_summary AS (
    SELECT
        job_field,
        donation
    FROM donation_data
)
SELECT
    job_field,
    SUM(donation) AS total_donation,
    COUNT(donation) AS number_of_donations
FROM job_field_donation_summary
GROUP BY job_field
ORDER BY total_donation DESC;
```

Total donation and number of donations above $200:

```
WITH donations_above_200 AS (
     SELECT
          donation
     FROM donation_data
     WHERE donation > 200
)
SELECT
     SUM(donation) AS total_donation,
     COUNT(*) AS number_of_donations
FROM donations_above_200;
```

Total donation and number of donations below $200:

```
WITH donations_below_200 AS (
     SELECT
          donation
     FROM donation_data
     WHERE donation < 200
)
SELECT
     SUM(donation) AS total_donation,
     COUNT(*) AS number_of_donations
FROM donations_below_200;
```

Top 10 states with the highest donations:

```
WITH state_donation_summary AS (
     SELECT
        state,
        donation
     FROM donation_data
)
SELECT
     state,
     SUM(donation) AS total_donation
FROM state_donation_summary
GROUP BY state
ORDER BY total_donation DESC
LIMIT 10;
```

Top 10 states with the least donations:

```
WITH state_donation_summary AS (
     SELECT
         state,
         donation
     FROM donation_data
)
SELECT
     state,
     SUM(donation) AS total_donation
FROM state_donation_summary
GROUP BY state
ORDER BY total_donation ASC
LIMIT 10;
```

Top 10 cars driven by the highest donors:

```
WITH donors_cars_table AS (
     SELECT
         donor_data.car,
         donation_data.donation
     FROM donor_data
     INNER JOIN donation_data
     ON donor_data.id = donation_data.id
)
SELECT
     car,
     SUM(donation) AS total_donation,
     COUNT(*) AS number_of_donors
FROM donors_cars_table
GROUP BY car
ORDER BY total_donation DESC
LIMIT 10;
```

Donor Ranking:

```
WITH donor_rankings AS (
     SELECT
         id,
         first_name,
         last_name,
         donation,
         RANK() OVER (
             ORDER BY donation DESC
         ) AS donor_rank
     FROM donation_data
)
SELECT *
FROM donor_rankings
ORDER BY donor_rank;
```

Average donation amount for each donation frequency:

```sql
WITH donation_frequency_analysis AS (
     SELECT
          donor_data.donation_frequency,
          donation_data.donation
     FROM donor_data
     INNER JOIN donation_data
     ON donor_data.id = donation_data.id
)
SELECT
     donation_frequency,
     COUNT(*) AS number_of_donations,
     SUM(donation) AS total_donation,
     ROUND(AVG(donation),2) AS average_donation
FROM donation_frequency_analysis
GROUP BY donation_frequency
ORDER BY average_donation DESC;
```

Highest donor in each state:

```sql
WITH ranked_donors AS (
     SELECT
          state,
          first_name,
          last_name,
          donation,
          RANK() OVER(
               PARTITION BY state
               ORDER BY donation DESC
          ) AS donor_rank
     FROM donation_data
)
SELECT
     state,
     first_name,
     last_name,
     donation
FROM ranked_donors
WHERE donor_rank = 1
ORDER BY donation DESC;
```

Supplementary Tables:

Highest donor by state:

State	First name	Last name	Donation
Michigan	Beverlie	Andriesse	$ 500.00
New York	Wallie	Leather	$ 500.00
Delaware	Peder	Rilton	$ 499.00
Virginia	Clevie	Camilletti	$ 499.00
Wisconsin	Worthy	Le feaver	$ 498.00
California	Tonnie	Stockney	$ 494.00
California	Corbett	Lansdale	$ 494.00
California	Nathaniel	McGenn	$ 494.00

Maryland	Beverlee	Camacke	$	493.00
Louisiana	Corbin	Rawne	$	493.00
Florida	Hurley	Bogey	$	492.00
Nevada	Emmit	McKenzie	$	491.00
New Mexico	Babbette	Fyers	$	491.00
Kentucky	Karilynn	Ivan	$	490.00
Arizona	Broderick	Dimitrijevic	$	489.00
Illinois	Padraig	Trittam	$	488.00
Utah	Patrick	Tardiff	$	487.00
Minnesota	Rupert	Hazelgreave	$	487.00
Texas	Read	Arboine	$	487.00
Texas	Karena	Andrieu	$	487.00
Texas	Tiffani	Tombleson	$	487.00
Washington	Ford	Evins	$	486.00
Arkansas	Nico	Twinterman	$	484.00
Pennsylvania	Wilma	Tummond	$	483.00
Connecticut	Emery	Rospars	$	483.00
Mississippi	Verene	Hearse	$	482.00
Massachusetts	Kellina	Eastcourt	$	482.00
Tennessee	Huntlee	Durbin	$	480.00
Indiana	Jeffrey	Lehmann	$	473.00
Georgia	Berenice	Kingsnorth	$	472.00
District of Columbia	Barbara-anne	Tonsley	$	470.00
Oregon	Saw	Guiraud	$	468.00
Missouri	Karoly	Gammon	$	463.00
Iowa	Kaitlin	Tomasicchio	$	461.00
New Jersey	Kacy	Cleall	$	460.00
Colorado	Olympie	Witherden	$	459.00
Ohio	Dana	Alster	$	452.00
Kansas	Rafferty	Brogini	$	440.00
New Hampshire	Reece	Paten	$	434.00
North Carolina	Jesus	Redfearn	$	423.00
Montana	Dar	Le Fleming	$	423.00
Nebraska	Jacobo	Wogan	$	422.00
Idaho	Marissa	Grimes	$	421.00
Oklahoma	Kanya	Pre	$	418.00
South Dakota	Betta	Moxom	$	401.00
North Dakota	Carly	Broxton	$	385.00
Alabama	Neysa	Palay	$	355.00
Alaska	Renaldo	Tuxill	$	332.00
Hawaii	Benny	Figure	$	320.00

Maine	Reginald	Ashburne	$	258.00
Wyoming	Adham	Imos	$	232.00
West Virginia	Jonell	Bonallack	$	184.00
South Carolina	Gannon	Ferries	$	175.00

Average donation by frequency:

Donation frequency	No. of donations	Donation		Average donation
Yearly	135	$	35,266.00	261.23
Seldom	118	$	30,650.00	259.75
Never	133	$	34,263.00	257.62
Often	111	$	28,476.00	256.54
Once	128	$	32,666.00	255.2
Weekly	129	$	31,645.00	245.31
Daily	128	$	29,249.00	228.51
Monthly	118	$	26,870.00	227.71

Dashboard KPI table:

METRIC	VALUES	
Total Donation	$	249,085.00
Total number of donations		1000
Average donation		249.09
Maximum donation	$	500.00
Minimum donation	$	5.00
Male total donation	$	127,628.00
Female total donation	$	121,457.00
Number of male donors		492
Number of female donors		508
Top donating state	California	

Entity Relationship Diagram: